Residential + Academic Support Platform (Hybrid SDLC)

A private multi-campus university is planning to modernize two major problem areas through a single integrated platform that supports both residential and academic operations. The first area is hostel operations, which are currently handled in a largely manual way. Room allocation is managed through paper forms and spreadsheets, and room changes often happen informally without a proper approval trail, creating confusion and disputes. There is frequent mismatch between finance and hostel records regarding student fee status, and maintenance/complaint requests are sometimes lost or delayed because they are written in logbooks rather than tracked in a formal system. Security entry/exit is also maintained manually, which makes it difficult to investigate incidents, verify visitor history, or produce reliable reports. Senior management has specifically asked for better reporting and audit trails across these hostel activities. In addition, the university has made it a firm requirement that prototypes must be built and validated with stakeholders before full development, especially for workflows involving room allocation, maintenance/complaints, fee verification, and security logging, so that usability and process correctness can be confirmed early.

The second area is attendance and academic monitoring, which is currently fragmented across manual attendance registers and scattered spreadsheets. This approach results in inconsistent records, delayed updates, disputes during audits, and it increases the chance of proxy attendance. Because the information is spread across departments and formats, the university has no reliable, unified view of attendance patterns, student engagement, or early warning indicators for students who may be at academic risk. To address this, the university wants a centralized attendance and monitoring system with role-based access, digital attendance capture, and a real-time student view that allows students to see their attendance status without waiting for end-of-term compilation. The system should also provide management and academic staff with dashboards and analytics, including risk alerts to identify low attendance trends and other engagement concerns. Over time, the university plans to extend the system with additional capabilities such as biometric or QR-based attendance, geofencing, integration with the LMS, and possibly parent notifications, but these are expected to evolve as the initial pilot is tested and refined.

A key constraint is that the university has a limited budget and requires a usable pilot quickly, while also anticipating that requirements will change after real users interact with the system in day-to-day operations. The platform must therefore serve multiple user groups and workflows, including students, wardens, maintenance staff, security staff, finance staff, faculty members, department administrators, and senior management, each with different needs, permissions, and expectations regarding accuracy, usability, and reporting.